

Nonhomogeneous Equations & Undetermined Coefficients



Complementary function and particular solution

- 1 For $y'' - 3y' + 2y = 4e^{3x}$, find the complementary function y_c .
 - 2 Explain why the general solution of a nonhomogeneous linear equation is $y = y_c + y_p$.
 - 3 For $y'' + 4y = 8$, find y_c and a constant particular solution y_p .
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Choosing a trial form (no resonance)

- 4 Find a particular solution of $y'' - 3y' + 2y = 4x$.
 - 5 Find a particular solution of $y'' + y' - 2y = 6e^{3x}$ (note $r = 3$ is not a root of $r^2 + r - 2 = 0$).
 - 6 Find a particular solution of $y'' - 4y = \cos x$.
 - 7 Find a particular solution of $y'' - y' - 2y = x^2$.
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The modification rule (resonance)

- 8 Find a particular solution of $y'' - 3y' + 2y = e^{2x}$. (Note $r = 2$ IS a root of $r^2 - 3r + 2$.)
 - 9 Find a particular solution of $y'' - 4y' + 4y = e^{2x}$. (Note $r = 2$ is a REPEATED root, so the usual modification xe^{2x} isn't enough.)
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Combining forcing terms

- 10 Find a particular solution of $y'' - y = e^x + x$. (Roots of $r^2 - 1 = 0$ are $r = \pm 1$, so e^x resonates but x does not.)
 - 11 Find a particular solution of $y'' + y = 3\sin x + 2$. (The forcing has both a resonant sine term and a non-resonant constant.)
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Solving full initial value problems

- 12 Solve $y'' - y = 4x$, $y(0) = 1$, $y'(0) = 1$.

- 13 Solve $y'' + y = 2\cos x$, $y(0) = 0$, $y'(0) = 0$. (This is a resonant forcing.)
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Application: a driven oscillator

- 14 A forced oscillator satisfies $y'' + 4y = \sin 3t$ (forcing frequency 3 $\neq$ natural frequency 2 — no resonance). Find a particular solution.
- 15 Now solve $y'' + 4y = \sin 2t$, where the forcing frequency EQUALS the natural frequency (resonance). Find a particular solution.
- 16 For the non-resonant case $y'' + 4y = \sin 3t$, use the particular solution above to state the full general solution.
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Choosing the correct trial form

- 17 State the correct (unmodified) trial form for a particular solution of $y'' - y' - 6y = 5e^{4x}$, and explain why no resonance check is needed here.
- 18 State the correct trial form for a particular solution of $y'' - y' - 6y = 5e^{3x}$, explaining why it must be modified.
- 19 State the correct trial form for a particular solution of $y'' + 9y = x\sin 3x$. (Note $\sin 3x$ resonates with the homogeneous solutions.)
- 20 State the correct trial form for a particular solution of $y'' - 2y' + y = 3xe^x$. (Note $r = 1$ is a repeated root of the homogeneous equation.)
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The method's limits

- 21 Explain why undetermined coefficients cannot be used directly on $y'' + y = \tan x$.
- 22 Explain why undetermined coefficients cannot be used directly on $y'' + y = \frac{1}{x}$, even though $\frac{1}{x}$ looks like a simple algebraic expression.
- 23 For $y'' - y = xe^x \ln x$ ($x > 0$), explain briefly why undetermined coefficients fails but variation of parameters still applies.
- 24 Is $y'' + y = \sec x$ (for $-\frac{\pi}{2} < x < \frac{\pi}{2}$) solvable by undetermined coefficients? Justify your answer.

- 25 Is $y'' - 4y = x^2 e^{3x} \cos x$ solvable by undetermined coefficients? If so, describe (without solving) the shape of the trial form.